Ukrainian Catholic University

AI Course · Semester Project Proposal · 2026

UA-Digest: Abstractive Summarization of Ukrainian News

Compact, factually accurate digests of Ukrainian news articles using a fine-tuned language model

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1. Problem and motivation

There are almost no tools for automatic news summarization in Ukrainian: available solutions either target English or use extractive methods that produce disjointed text. Ukrainian media publish hundreds of articles a day, and readers lack a quick way to get a compact, factually accurate digest. Existing browser extensions for other languages rely on extractive sentence selection, which tends to string together grammatically disconnected fragments and loses the narrative flow that readers expect from a lead paragraph. The project has practical value (a plugin for reading news) and research value (evaluating the quality of abstractive summarization for a low-resource, morphologically rich language where labeled summarization data is scarce).

2. Related work

Abstractive summarization is based on seq2seq architectures with an attention mechanism [1]. BERT-based models and their distilled versions [2, 3] show strong results for English but require adaptation for the morphologically rich Ukrainian language: subword tokenizers trained on English corpora fragment Ukrainian words far more aggressively, which hurts both training efficiency and generation fluency. ROUGE [4] remains the standard metric for evaluating summary quality, which we will supplement with a manual expert assessment of factual accuracy, since ROUGE alone does not penalize hallucinated facts that are lexically similar to the source text.

3. Data

The primary dataset is an open corpus of Ukrainian literary and journalistic prose [5], supplemented with a manually collected set of ~8 000 "article–headline + lead" pairs from three public RSS feeds of Ukrainian news outlets (usage license to be confirmed with rights holders; for training we will use only articles under an open license or with permission for research use). We will deduplicate near-identical articles (syndicated wire content republished by multiple outlets) using MinHash on word shingles, and filter out articles shorter than 200 words, since these rarely contain enough content for a meaningful lead. Split: 80 / 10 / 10% (train/val/test), stratified by outlet to avoid a single source dominating any one split.

4. Approach and methods

The base model is mT5-small, fine-tuned on the collected corpus using teacher forcing with a cross-entropy loss. We plan to compare two configurations: (1) direct fine-tuning of mT5-small, (2) mT5-small with an additional self-supervised adaptation stage on an unannotated Ukrainian corpus before fine-tuning. We use the Hugging Face Transformers library implementation as a base, adapting tokenization and the sampling scheme for Ukrainian. Training will use mixed-precision on a single GPU with gradient accumulation to reach an effective batch size of 32, early stopping on validation ROUGE-L, and beam search (beam size 4) at inference time to reduce repetition artifacts common in small fine-tuned models.

5. Evaluation

Metrics: ROUGE-1/2/L [4] against reference leads. Baseline: extractive LexRank and a plain first-two-sentences cutoff ("lead-2"). Additionally, a manual evaluation of 50 random summaries by three independent assessors on a 1–5 scale (coherence, factual accuracy). We expect to beat the lead-2 baseline by at least 15% on ROUGE-L and achieve an average factual accuracy score of $\geq 4/5$. We will also report inter-annotator agreement (Krippendorff's alpha) for the manual scores, and inspect a sample of low-scoring outputs by hand to categorize the dominant failure modes (e.g. hallucinated numbers vs. dropped named entities).

6. Plan and risks

> **Key numbers:** ~8 000 article pairs · 2 model configurations · 6 weeks to midterm

Before midterm (6 weeks): data collection and cleaning, basic mT5-small fine-tuning, first ROUGE run against the baseline. **Before the final (6 more weeks):** second model configuration, manual evaluation, a simple web deployment demo. **Risks:** (1) not enough high-quality annotated pairs – plan B: expand the corpus with semi-automatic lead generation; (2) low summary quality from mT5-small – plan B: switch to a distilled but larger model (mT5-base) if compute resources allow; (3) rights holders decline permission for a subset of the RSS-collected articles – plan B: fall back to the open literary/journalistic corpus alone and treat the RSS set as an optional held-out evaluation sample rather than training data.

7. Contribution statement

- **Ivan Petrenko** — 35%: data collection and preprocessing, EDA, manual annotation
- **Olena Bondarenko** — 40%: model fine-tuning, experiments, hyperparameter tuning
- **Andrii Kovalenko** — 25%: training infrastructure, web deployment demo, documentation

References

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[4] Chin-Yew Lin. “ROUGE: A Package for Automatic Evaluation of Summaries.” In: *Text Summarization Branches Out*. 2004. URL: https://aclanthology.org/W04-1013.

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